University of BRISTOL

AWS INDUCTION

Introduction to AWS Cloud Services available at UoB

bristol.ac.uk



What is AWS?

- Cloud computing = the **on-demand** delivery of IT resources with **pay-as-you-go** pricing.
- Cloud services can **grow as you need** them to – you pay for what you need as you need it by **dynamically assigning resources.**
- Every action that you make in AWS is an API call that is authenticated and authorized.
- Administered via: Management Console, Command Line and SDK tools.
- **Security: 'Shared Responsibility Model'** You are responsible for security '**in**' the cloud, AWS responsible for security '**of**' the cloud. NB - this may vary from service to service!



Common AWS Services

(Too many to cover!)

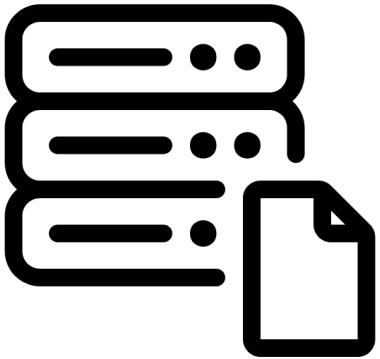
Essentially three types of compute platform:

- **Virtual Machines** (VMs) e.g. Elastic Compute Cloud (EC2) - virtual servers, tools for dynamic scaling, GPU access, used to host traditional apps with full access to OS
- **Container** (e.g. Docker) **Services** - Container tools like ECS and EKS, can run on EC2 or managed on 'Fargate' service.
- **Serverless** – Managed by AWS, no OS level access e.g. Lambda can host short running functions (under 15 mins) in response to triggers.



COMPARE COMPUTE SERVICES:
https://docs.aws.amazon.com/decision-guides/latest/compute-on-aws-how-to-choose/choosing-aws-compute-service.html

Storage

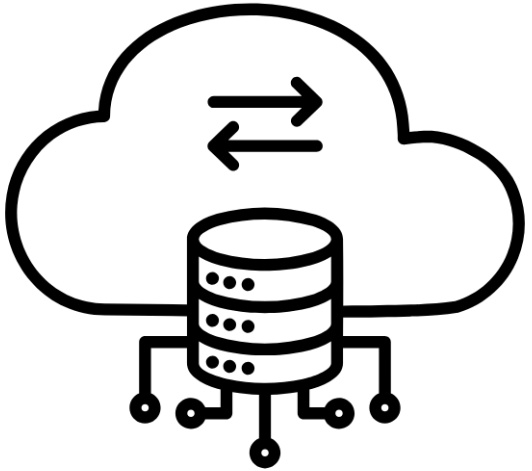


- **Instance store volumes** – Free VM storage volumes for **temporary** data that is deleted when you stop, hibernate, or terminate your VM instance.
- **Elastic Block Storage (EBS)** volumes – Chargeable, **persistent**, low latency, block storage designed to be directly attached to EC2 instances.
- **Simple Storage Service (S3 'buckets')** – Chargeable, data stored as objects with associated metadata, offering scalability, durability, and availability.

OTHER TYPES OF STORAGE ARE AVAILABLE, COMPARE HERE:
https://docs.aws.amazon.com/decision-guides/latest/storage-on-aws-how-to-choose/choosing-aws-storage-service.html

Database services

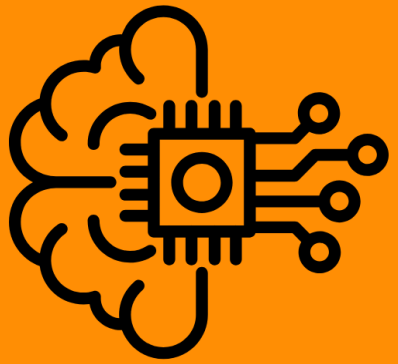
- **Amazon Relational Database Service** (Amazon RDS) - **relational** database service, provides cost-efficient, resizable capacity for an industry-standard relational database and manages common database administration tasks.
- **Amazon DynamoDB** - serverless, **NoSQL**, fully managed database with single-digit millisecond performance at any scale. Aims to address scaling and operational complexities of relational databases.



CHOOSING AN AWS DATABASE SERVICE:
https://docs.aws.amazon.com/decision-guides/latest/databases-on-aws-how-to-choose/databases-on-aws-how-to-choose.html

Machine Learning and AI

- **Amazon Sagemaker** – fully managed service designed to help you **build, train, and deploy** machine learning models at scale.
- **Amazon Bedrock - discover, test, and use** over 100 popular, emerging, and specialized foundation models (FMs).
- You do not need Sagemaker to use EC2 GPUs or CPUs with a Deep Learning AMI.

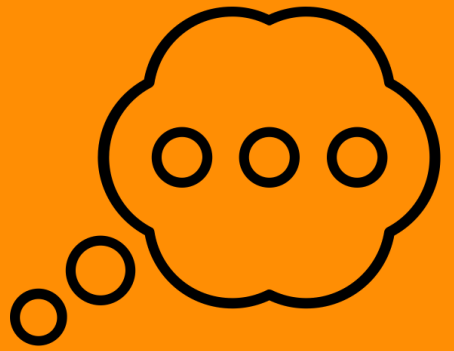


COMPARISON BETWEEN BEDROCK AND SAGEMAKER:
https://docs.aws.amazon.com/decision-guides/latest/bedrock-or-sagemaker/bedrock-or-sagemaker.html

Do you need AWS?

Alternatives:

- Windows virtual Desktop
  - Open access
  - Engineering with GPU
- Remote Computer Labs
- Self Service Cloud
- BlueCrystal and BluePebble HPC



Windows Virtual Desktop

- Remote desktop onto University network
- Access specialist software
- Open Access – same environment as open access computing labs on campus
- Engineering with GPU – access to Eng build environment with GPU
- Can not be left logged in for long periods!

Remote Computer Labs

- Remote desktop to specific physical computers on-campus
- Limited capacity
- Can not be left logged in for long periods!

MORE INFO:
https://uob.sharepoint.com/sites/itservices/SitePages/student-remote-desktop.aspx

Self Service Cloud

- Linux Red Hat 8 or Windows Server 2019
- No backup
- Requires an administrator to be set up to manage VMs



MORE INFO:
https://uob.sharepoint.com/sites/itservices/SitePages/SelfServiceCloud/self-servicecloud.aspx

BlueCrystal and BluePebble HPC

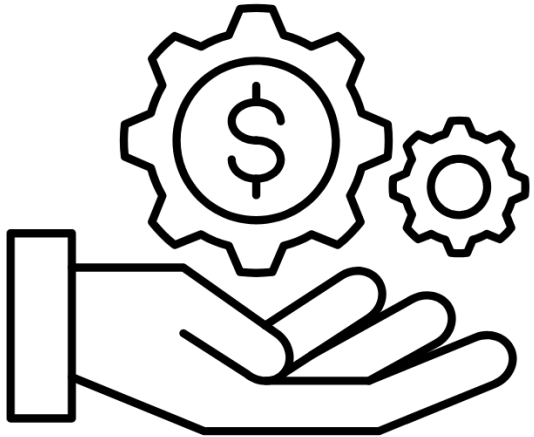
- For taught students, must be requested by teaching staff for specific course with a 6-week lead time
- For PG research, supervisor must apply

MORE INFO:
https://www.bristol.ac.uk/acrc/high-performance-computing/

High Performance Computing



With great power comes great responsibility…



Please consider carefully:

- Do the research and ask for support from your supervisor, the Tech Hub or the Digital Team
- Beware of data breaches
- Don’t store sensitive data – consider using synthesised data if possible
- Respect GDPR
- Avoid cost overruns – deallocate resources when you are not using them!
- Respect all relevant university policies for acceptable use and information security
- No personal gain
- Be responsive to emails from us and AWS
- Have you done an ethics review?

Security and permissions

- Identity and Access Management (IAM) used for authentication and authorisation.
- IAM policies take care of permissions - authorisation.
- IAM permission policies are JSONs.
- IAM Roles - credentials that can be assigned to IAM users or services, like an EC2 instance or a Lambda function. Roles are assigned programmatically, and authorisation is invoked at point of use and revoked after



CHECK OUT 'MODULE 9 – SECURITY' ON 'AWS Cloud Practitioner Essentials' TRAINING

Security best practices



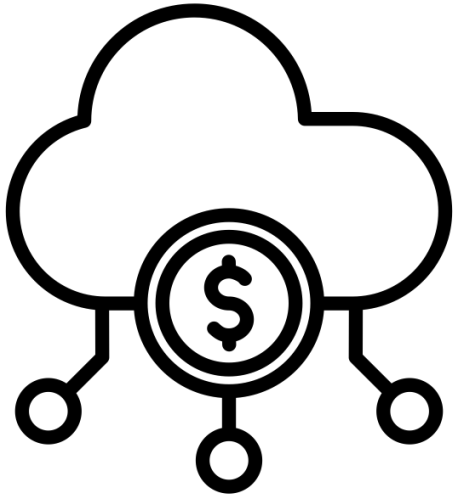
- Do you need to be internet facing?
- Use **IP whitelist/blacklisting**
- Use **VPC (virtual private cloud)** and associated network hardening tools
- Use **Security Groups/Route Tables**
- Keep your login credentials secure, **do not share**!
- Assign permissions with **least privilege** required
- Respect University policy security practices
- Secure your account from day 1!

CHECK OUT 'MODULE 9 – SECURITY' ON 'AWS Cloud Practitioner Essentials' TRAINING

Cost Awareness

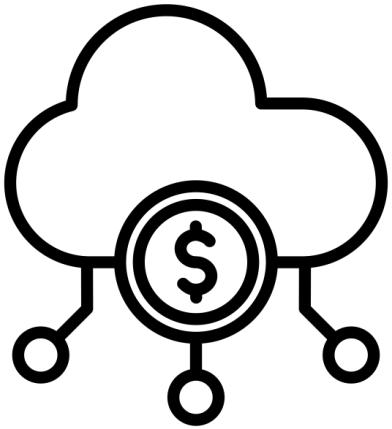
Make use of the cloud stuff!

- Dynamically **start and stop services** as required, don't leave them running.
- Do you need a whole VM instance, or will a Lambda function do what you want?
- 'Spot instances' make **90%** saving if you can deal with potential interruption!
- Would your application be more cost-effective running as a container, perhaps on a serverless platform?



INTRODUCTION TO AWS PRICING
https://skillbuilder.aws/learn/94T2BEN85A/aws-cloud-practitioner-essentials/8D79F3AVR7

Cost Awareness



- Make sure you use the correct type and size of VM
- Configure auto-scaling correctly
- Beware of setting minimum instances
- Regularly check and terminate any unused services or instances
- Be aware not to leave paid services running, e.g. EC2 instances, EC2 elastic IPs, S3, RDS/Dynamo DB services, VPC, beware of autoscaling and minimum requirements.
- Stay in the EU region
- Use **AWS cost monitoring features** and alerts to stay informed about your spending and resource usage.

USE THE 'AWS Cost Explorer' TO SEE HOW MUCH YOU ARE SPENDING

Your budget

- Start with $100 budget, if you think you will need more then discuss with your supervisor and contact the Tech Hub.
- Use AWS cost monitoring features and alerts to stay informed about your spending and resource usage.
- Your budget is not a target!



USE THE 'AWS Pricing Calculator' TO WORK OUT WHAT YOU MIGHT SPEND OVER THE DURATION OF YOUR PROJECT.

Running out of budget?

1. You (and we) will receive email alerts when your costs have reached 50%/75%/95% of the budget.
2. Contact the Tech Hub as soon as possible
3. If you go over budget, we will contact you and may suspend your services
4. You will need an agreement from your supervisor to extend your budget

USE THE 'AWS Cost Explorer' TO SEE HOW MUCH YOU ARE SPENDING AND WATCH YOUR EMAILS!

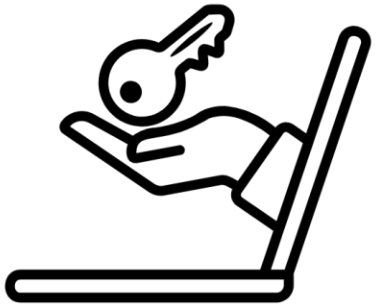


What services are available?

You will be given access to almost all AWS services (i.e. EC2, S3, Lambda, RDS, DynamoDB, etc.), but you will need to specifically request the following:

- **Sagemaker/Sagemaker Studio** - you will need written approval from your supervisor as this can be costly. You do not need Sagemaker to use EC2 GPUs or CPUs with a Deep Learning AMI.
- **GPUs** - regions need to be validated, and the default vCPU quota is 0. You can request an increase through the service quota page in your account. EU regions are easier/quicker to set up and we would prefer if you could stick to London, Stockholm, or Ireland.

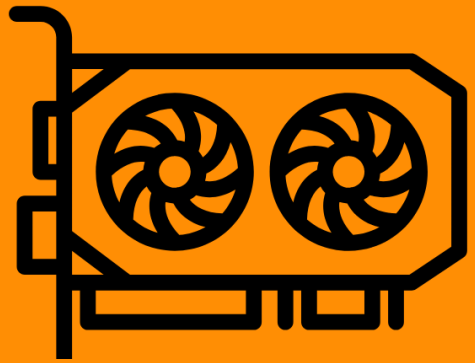
AWS IMPLEMENT DEFAULT SECURITY MEASURES TO PREVENT THE ABUSE OF GPU INSTANCES (E.G. CRYPTO MINING)



Types of GPU instance

- P family – high-performance, general-purpose computing and machine learning
- G family – graphics-intensive applications and machine learning inference

There can be a big price difference between a ‘small’ and a ‘large’. Go for the minimum you think you will need, and we can top it up if we need to.



Regions



- Stay in the EU region whenever possible, it is required for GDPR compliance.

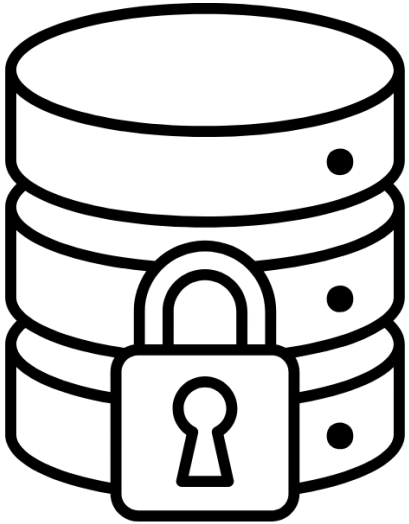
Otherwise:

- Choose regions based on compliance, proximity, feature availability, pricing

THERE MAY BE COST DIFFERENCES BETWEEN REGIONS!!
Choose your region based on cost

Data protection

- Can you use synthetic data?
- Respect GDPR, you are the ‘Data Controller’! It may be necessary to stay in EU region to comply.
- Respect University Policies:
  Information Security Policy
- Have you performed an Ethics review?



Backing up

- Implement regular backups for critical data and source code, particularly for work that cannot easily be reproduced.
- Utilise version control (e.g. GitLab) for your projects wherever possible, ensuring you maintain local copies or separate repositories to prevent data loss.
- Remember VM instance storage is temporary!!



Compliance and institutional policy

You must comply with University guidelines and policies, or access will be removed.

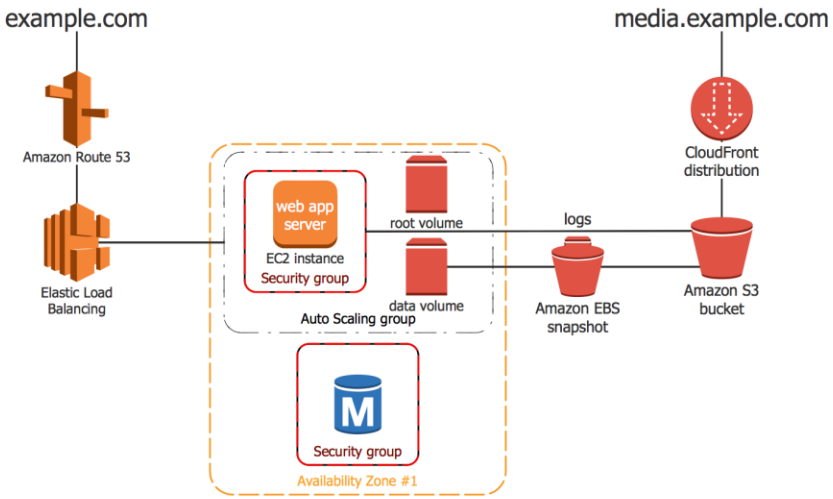


University of Bristol Policies

- Digital Services Team Policy
- Acceptable use Policy
- Information Security Policy
- Code of Conduct for Computing Facilities
- Research Data Management and Open Data Policy

Common scenarios and templates

Example Web App Architecture



Example Cloudformation Template - To launch an instance from a launch template

```
Resources:
  myInstance:
    Type: 'AWS::EC2::Instance'
    Properties:
      LaunchTemplate:
        LaunchTemplateId: !Ref myLaunchTemplate
        Version: !GetAtt myLaunchTemplate.DefaultVersionNumber
  myLaunchTemplate:
    Type: 'AWS::EC2::LaunchTemplate'
    Properties:
      LaunchTemplateData:
        ImageId: ami-0a70b9d193ae8a799
        InstanceType: t2.micro
        MetadataOptions:
          InstanceMetadataTags: enabled
        SecurityGroupIds:
          - sg-12a4c434
        TagSpecifications:
          - ResourceType: instance
            Tags:
            - Key: department
              Value: dev
```

Other Cloudformation templates for all AWS services can be found here: https://docs.aws.amazon.com/AWSCloudFormation/latest/TemplateReference/aws-template-resource-type-ref.html

How to shut everything down in emergencies

Don’t Panic!

Go to the Tech Hub or call on
+44 117 455 0461



bristol.ac.uk

Cleanup and citizenship

- Please **notify the Tech-Hub when you are finished** with your account. We will keep it alive until marking is complete but can suspend services and stop incurring cost.
- **Do the research** – work out what services are best for your application and the most economical, work out what your costs will be before you start.
- **Only use what you need** - you will save energy and money. Go for the minimum and increase as necessary.
- **Deallocate** resources when you are not using them.
- Be **responsive to emails** from us and **alerts** from AWS.

Your account may be frozen if you do not respond to emails within 7 days!!



LEARN: AWS Well Architected

Additional training resources/information

- If in doubt, or you have any questions, get in touch with the Tech Hub:
  engf-tech-hub@bristol.ac.uk
- If you anticipate needing additional budget, Sagemaker or GPU, consult with your supervisor first.

'AWS Cloud Practitioner Essentials' **TRAINING, FIND THE SPECIFIC MODULE FOR WHAT YOU WANT TO KNOW ABOUT.**

'AWS Decision Guides' **TO COMPARE SERVICES.**

'AWS Cost Explorer' TO SEE HOW MUCH YOU ARE SPENDING.

'AWS Pricing Calculator' **TO WORK OUT WHAT YOU MIGHT SPEND.**

University of BRISTOL

CONGRATULATIONS!

- You have now completed the AWS Induction follow this link to request an account.
- Default access is automated and will be done within 2 working days of the receipt of request
- If you want GPU, Sagemaker, AI/ML, then this will need human review and could take up to week.



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